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Q1a)

The crab's data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two-colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

Q1b) Data structure of Data set Crabs from Package MASS

```
str(MYDATA1)

'data.frame':  200 obs. of  8 variables:
 $ sp      : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex     : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index: int   1 2 3 4 5 6 7 8 9 10 ...
 $ FL      : num  8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW      : num  6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL      : num  16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW      : num  19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD      : num   7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

Q1c Frequency distribution for species and sex

variable	frequency
Specie-blue	100
Specie-orange	100
Female	100
male	100

```
ctab1<-table(MYDATA1$sp,MYDATA1$sex)

barplot(ctab1,beside = T,legend=T,col =
rainbow(4),lwd=2,
```

Q1d) barplot showing the distribution species and sex status

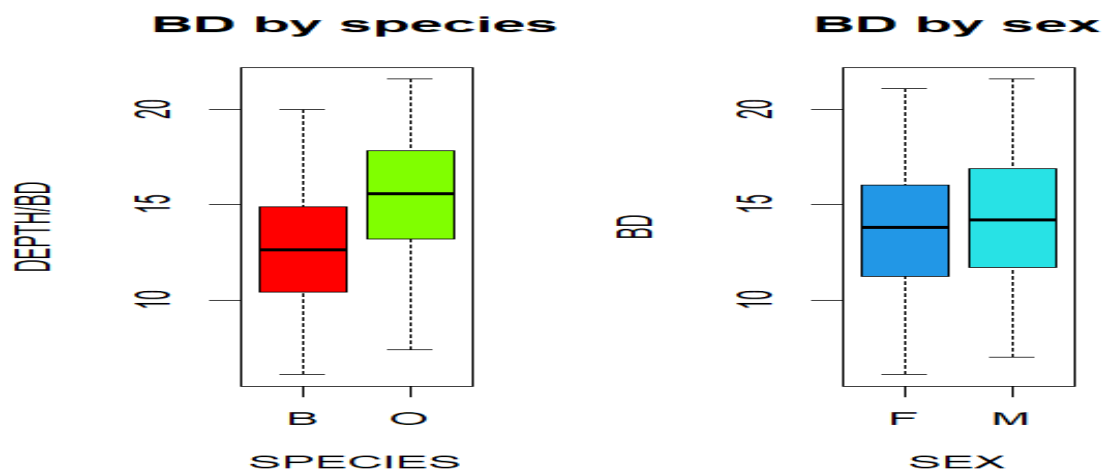


1Qe: Balanced

Since all the groups have equal number of frequencies (100 observation each), we can infer that the data is balance across the four groups.

Question2

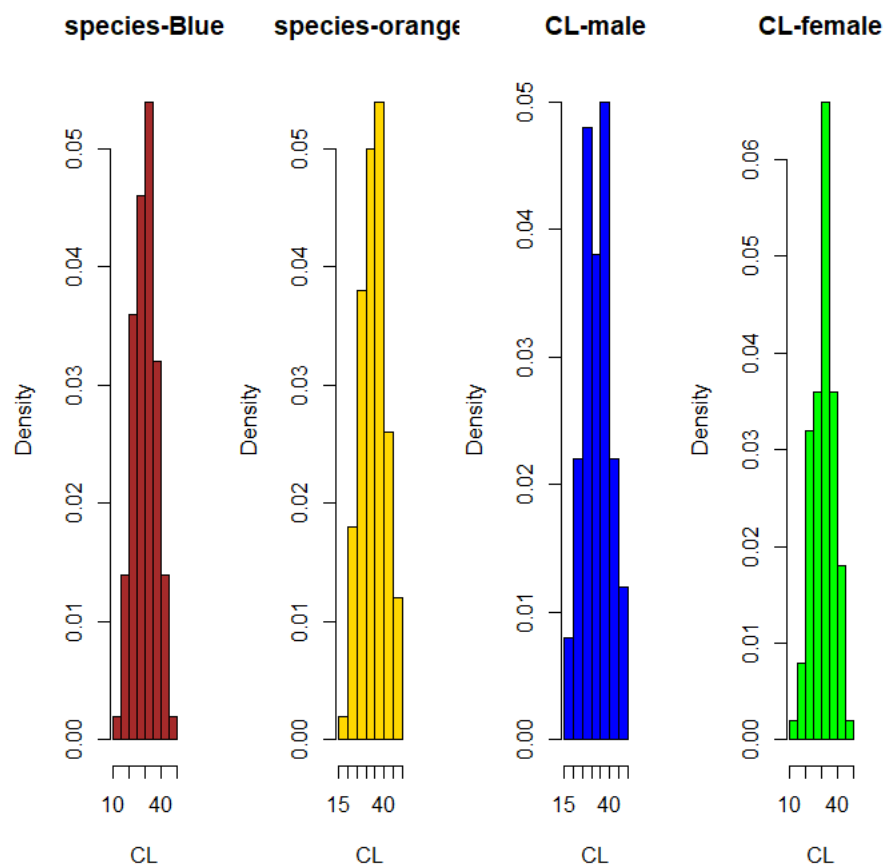
Q2a) side by side boxplot of body dept (BD) for the four groups(sex and species)



b) There were no outliers of body dept between the four groups. Also, all the categories in each group appear to be normally distributed as their first quarters and their 3Qur have the same shape as illustrated from the above boxplot. The median values divide the 1QUr and the 3Qur

into equal parts across the four groups. Hence, body dept appear to be normally distributed across the four species. However, the median body dept of specie orange is greater than species blue. For the case of sex, the median body dept of male is quite higher (but can't justify its significance). Variability in blue and female body dept are also quite greater than orange and male respectively.

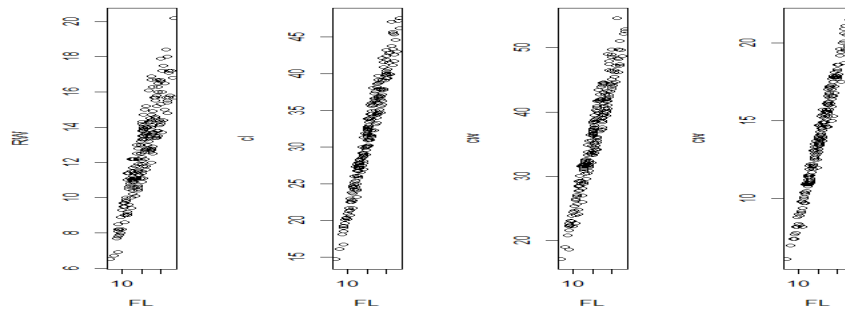
Q3a) histogram showing the distribution of carapace length (CL) for each species



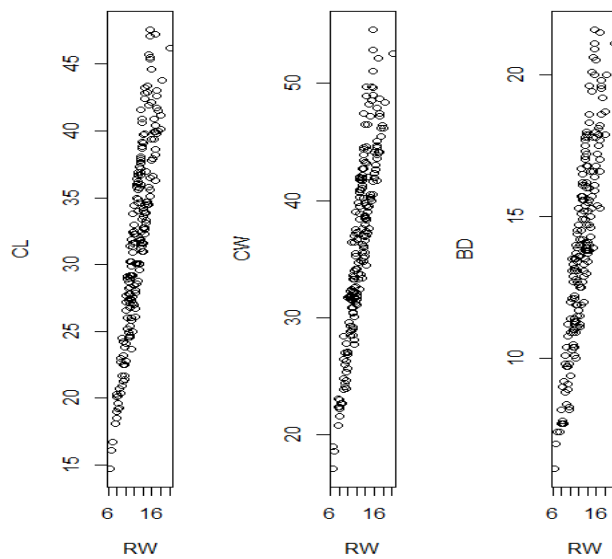
Q3b)

Carapace length in the blue species appears to be quite normally distributed by slight positively skewed with skewness value 0.03(~ 0). The modal value of CL in blue is 27.3 which is different from its mean and median values. For orange, the modal CL is 31.6 which is different from its mean and median. Its not normally distributed since its skewness is -0.15(negatively). but visually appears normal. For male, the skewness is -0.08 and modal value of 29.2. Hence not normally distributed. For female, negatively skewed(-0.13) and mode of 34.1 which is different its mean and median. Hence CL IS not normally distributed in females.

Q5a) relationship between FL and on other measurement(RW,CL,CW AND bd)



Relationship between CL and on other measurement(CW AND bd)



Q5b correlation output

```
cor(MYDATA1[4:8])
```

	FL	RW	CL	CW	BD
FL	1.0000000	0.9069876	0.9788418	0.9649558	0.9876272
RW	0.9069876	1.0000000	0.8927430	0.9004021	0.8892054
CL	0.9788418	0.8927430	1.0000000	0.9950225	0.9832038
CW	0.9649558	0.9004021	0.9950225	1.0000000	0.9678117
BD	0.9876272	0.8892054	0.9832038	0.9678117	1.0000000

There is strong positive relationship among variables.

Q6)

The dataset crabs is a data frame with 8 variables. Two are categorical and the rest being numeric. There is a strong positive relationship among the variables used for the measurement. CL is not normally distributed among the four group.